Ocean Sound Separation with Causal High-Resolution Background Priors

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***Abstract*—We test whether a causal background prior from the preceding ocean soundscape improves separation of an event in the next five seconds. DCLDE mixtures pair real annotation-clean backgrounds with synthetic targets, providing exact stems for controlled evaluation. A failed 30-band FiLM precursor motivates an information ladder and a time-resolved 128-group background map. We inject this map and an explicit mixture-minus-background innovation through a residual BS-RoFormer adapter. On 4,096 matched test mixtures, three separator seeds average 13.912 dB target SI-SDRi with past-60-s Persistence, versus 11.511 for fixed-mean Zero, 11.767 for wrong-day Shuffled, and 9.635 for ordinary BS-RoFormer. *Persistence improves over Zero and Shuffled by* $2.400 \pm .237$ *and* $2.145 \pm .181$ *dB, respectively, on all nine seed-day pairs; a learned GRU does not consistently improve on Persistence* ($+.017 \pm .036$ **dB across seeds). A single-seed true-current-background Oracle reaches 16.767 dB. On 373 previously unseen Gulf of St. Lawrence calls, correct history lowers the frozen detector's background-stem response relative to wrong-file conditioning, but slightly worsens event retention and off-event preservation. These real-event proxies indicate partial transfer; isolated whale stems are unavailable.**

***Index Terms*—ocean acoustics, source separation, causal background prior, BS-RoFormer, weak-reference evaluation**

I. Introduction

Ocean backgrounds vary with shipping, weather, biology, and season [1]. A mixture-only separator must infer both an unexpected event and this changing background from the same observation. We instead ask whether the preceding soundscape can supply a causal prior. Our supervised data tuple and main model are

$$\begin{aligned} &x_t = N_t + s_t, \qquad \widehat{B}_t = P(H_t), \\ &\widehat{R}_t = \log|X_t| - \widehat{B}_t, \\ &[X_t, \widehat{B}_t, \widehat{R}_t] \xrightarrow{\text{Residual+BSR}} (\hat{s}_t, \hat{N}_t), \end{aligned} \tag{1}$$

where H_t excludes annotated call intervals and N_t is a real current no-call segment. Training has exact stems, while the separator never receives N_t or s_t as input. Here X_t denotes the current mixture STFT and $\widehat{B}_t$ is a causal background log-magnitude map; P is either last-map Persistence or a learned predictor. The 30-band FiLM precursor failed on held-out audio, motivating an information ladder and the 128-group residual interface in Fig. 1.

Time-domain separators [2], [3], attention models [4], [5], and band-split systems [6], [7] provide strong mixture-only backbones. Hybrid Transformers and MixIT broaden the architecture and supervision regimes [8], [9], but do not test a forecast of the upcoming background. FiLM, cross-attention, and cross-modal separation [10]–[12] motivate three interfaces. Forecasting baselines range from recurrent to linear and patch-based models [13]–[16].

Our contributions follow one chain: (1) an information ladder explaining the failed coarse condition; (2) annotation-clean real backgrounds and masked histories feeding a 128-group residual interface;[1] (3) matched Zero, Shuffled, Persistence, Forecast, and Oracle controls with three separator seeds; and (4) a frozen-model test at an untouched recording site. Real-event adaptation studies diagnose transfer limits and are secondary to the full-reference result.

II. Method

A. High-resolution causal residual separator

The main history $H_t \in \mathbb{R}^{12\times 79\times 128}$ contains twelve strictly prior 5-s, 128-group log-magnitude maps. A 512-point Hann STFT with 128-point hop gives 79 frames per window at 2 kHz. Annotated event windows are masked and filled only with nearby clean windows within the past 60 s. The main causal reference is last-map Persistence, $\widehat{B}_t = B_{t-1}$. A comparison arm uses a residual GRU to compress each map into four temporal chunks and predict the next full 79×128 map; its zero-initialized decoder starts at Persistence. The GRU is trained on clean training days, selected by validation map MAE, and frozen before separator training.

We form $\widehat{R}_t = \log|X_t| - \widehat{B}_t$ on the same grid. The time-aligned pair $[\widehat{B}_t, \widehat{R}_t]$ is projected into mixture-derived BSR band tokens before axial attention. The residual projection starts at zero, so every conditioned separator initially equals the same transferred mixture-only BSR waveform mapping. We train with exact synthetic event stems and a waveform plus multi-resolution STFT loss. Figure 1 shows the data and inference paths.

B. Preliminary 30-band FiLM design

The precursor summarized each of twelve preceding 5-s clean windows into 30 log-power bands and used a residual GRU to predict the next summary. FiLM scaled and shifted BS-RoFormer band tokens using this prediction. Matched controls replaced it with zero, a wrong-day summary, persistence, or the true current $E(N_t)$. This *30-band Oracle-FiLM* still discards within-window time structure and phase; it is not an upper bound for a richer background representation. We retain this failed design only to motivate the information ladder below.

[1]Code and reproduction instructions: https://github.com/D-eval/ocean_noise_pred_splite.

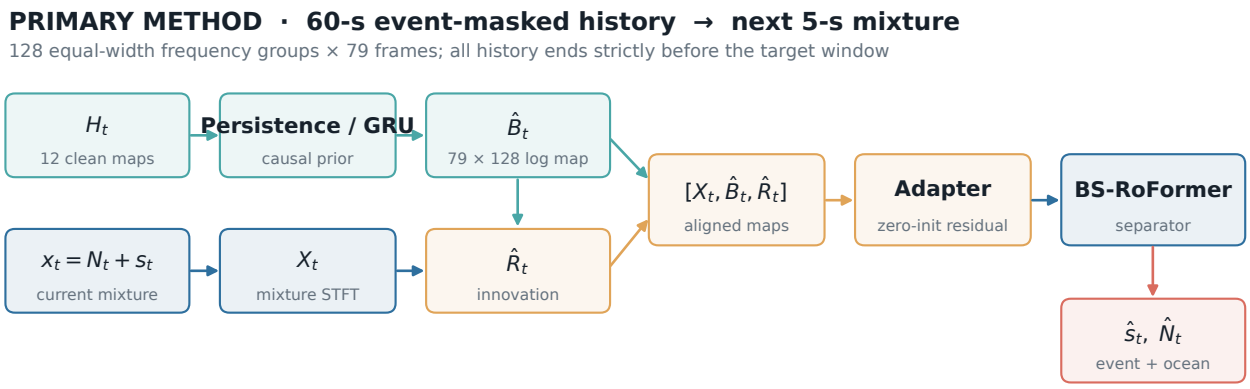


Fig. 1. Main 128-group causal residual architecture. The event-masked 60-s history supplies a future 5-s background map by Persistence or GRU; the explicit innovation is $\widehat{R}_t = \log|X_t| - \widehat{B}_t$. The same residual adapter is used for Zero, wrong-day Shuffled, Persistence, Forecast, and current-background Oracle conditions. Only the Oracle sees current clean background as an input.

C. Marine-aware source distribution

The broad generator contains 13 families: harmonic, linear chirp, inharmonic, impulse train, frequency-modulated tone, band noise, whale upsweep, downsweep, arch, S-curve, pulse packet, vessel tonal, and vessel cavitation. Whale proxies use one to three localized events with variable duration, curvature, partials, taper, and flutter. Vessel-tonal and narrow-band cavitation components draw a 5–20 Hz low-frequency-oscillator amplitude envelope with probability 0.8; their broadband component is not modulated. Training SNR is uniform on $[-18, 12]$ dB. This distribution is intended to cover contour and modulation patterns, not to claim species identity.

We quantify spectral displacement with directional $D_{\mathrm{KL}}(p\|q)$ and finite symmetric Jensen–Shannon (JS) divergence [17]; spectral entropy and source–ocean overlap are reported separately. Larger KL/JS expands coverage but is not treated as semantic diversity. The DCLDE main experiment samples only its five whale and two vessel families; all six main-domain models see the same target distribution.

D. Main data construction and secondary ablations

The main dataset uses day-disjoint 2-kHz DCLDE 2013 recordings and whale-call annotations [18], [19]. Across 168 h of WAV, excluding annotated calls and a 2-s guard on each side yields 146.4 h of annotation-clean 5-s windows. For each target, the 12 immediately preceding 5-s blocks from the same WAV form a high-resolution history. Annotated calls and their 2-s guards are masked; 235 of 68,075 eligible current windows have one masked historical block, filled by the nearest clean block still within the preceding 60 s. Histories never cross WAV boundaries or use the current window. This leaves 94.5 h eligible overall, split into 37.3/9.7/47.6 h for train/validation/test. For each current clean N_t, a seven-family whale/vessel synthetic event s_t is mixed to form $x_t = N_t + s_t$. The main split is by recording day, and the GRU receives only event-masked features from the immediately preceding 5-s windows.

Secondary ablations use NOAA–Navy SanctSound hourly TOL products [20]. MB03 deployments 01–02/03/04 are train/validation/test. Random-phase inversion creates controlled spectral surrogates, not recovered audio. Concatenation, cross-attention, KL/JS coverage, and the one-hour history-resolution comparison are confined to this domain. The separate 42-h SanctSound WAV strength sweep also lacks event annotations, so it is a useful controlled real-background ablation but is not counted as clean-background primary evidence.

Complete-reference tiers report SI-SDR and SI-SDRi using scale-invariant projection [21], [22]. All uncertainty intervals are seeded paired bootstraps, blocked by recording day for the main test. The event classifier is frozen before any real-event evaluation. Current test features are used only as targets for the synthetic full-reference task, never as GRU input; all histories end before t.

III. Experiments and Results

A. Preliminary conditioning-mode ablation

The initial 30-band design selected FiLM. Table I compares it with alternative interfaces on controlled 5-s spectral-surrogate examples; on 832 fixed-0-dB pairs (64 per family), FiLM improves 4.850 to 5.541 dB SI-SDRi. Band-wise scale and bias directly modulate existing band evidence. Concatenation must compress doubled features, while cross-attention must learn alignment from coarse predicted spectra under the same budget.

FiLM gains 0.869, 0.835, 0.919, 0.906, and 0.806 dB on the five whale proxy families. Vessel tonal is the hardest family: the conditioned absolute SI-SDRi is -8.295 dB, despite a 0.320-dB relative gain. Overall gain has weak correlation with JS

TABLE I
CONDITIONING-MODE ABLATION ON TOL-DERIVED SPECTRAL SURROGATES (DB).

Injection	Val. SI-SDRi	Test SI-SDRi	Paired Δ_0
None	4.864	4.059	0
Concatenation	5.343	4.432	+0.623
FiLM	**5.419**	**4.468**	**+0.692**
Cross-attention	5.128	4.261	+0.331

TABLE II
PRELIMINARY 30-BAND FILM AUDIT. TEST TARGET SI-SDRI (DB) ON THE SAME 4,096 DCLDE MIXTURES; THE ORACLE SEES ONLY $E(N_t)$.

Condition	Test SI-SDRi
BS-RoFormer / Zero-FiLM	9.631 / 9.631
Shuffled / Persistence-FiLM	9.526 / 9.524
GRU-Forecast / Oracle-FiLM	9.497 / 9.535

($r = -.057$) and entropy ($r = .122$), so increasing divergence broadens support but does not predict usefulness.

B. Preliminary 30-band diagnosis

We repaired an earlier baseline label by training a truly injector-bypassed BS-RoFormer and retaining the mislabeled checkpoint as Zero-FiLM. All six 30-band conditions use matched DCLDE mixtures, initialization, training and selection rules, and 4,096 examples from three held-out days. None improves on BSR (Table II); even the current-background Oracle sees only the compressed $E(N_t)$ and scores 9.535 versus 9.631 dB. The GRU forecast's lower validation MAE (1.926 versus 2.111 dB for persistence) therefore does not establish a separation gain.

a) Full-waveform copy sanity check.: With the same BSR frozen, a fitted three-coefficient head receives BSR output, mixture, and a condition waveform. When given the actual N_t, it learns $\hat{s}_t = x_t - N_t$ to numerical precision (80.755 dB SI-SDRi); zero and wrong-time waveforms score 9.231 dB. This noncausal check verifies an answer path, not the value of 30-band FiLM or a causal predictor.

b) Oracle information ladder.: We next reduce the information in the *current* N_t while keeping the frozen BSR, the three-coefficient head, training examples, and 4,096 matched test mixtures fixed. A 512-point Hann STFT at 2 kHz has 257 frequency bins on a 64-ms hop. We average background power across either adjacent equal-width groups (128, 64, 32) or geometric 20–950-Hz groups (128, 64, 48, 30), and separately pool frames over 0.25, 1, or 5 s. The retained magnitude is inverted with either the actual N_t phase or the mixture x_t phase. Geometric groups absorb edge bins; only 95 of the nominal 128 geometric groups contain an STFT bin. The 30-band/5-s rung is analogous to, but not numerically identical with, the original $E(N_t)$.

Figure 2 identifies where this *copy path* ceases to show a held-out gain. Exact amplitude with mixture phase still adds 11.164 dB over BSR (day-block CI $[10.889, 11.594]$). With equal-width groups, 128 retains $+2.067$ dB ($[0.969, 3.386]$), while 64 has -0.623 dB ($[-1.404, 0.259]$): the confirmed gain disappears between these levels. For geometric bands, nominal 128 already gives -0.529 dB ($[-1.257, 0.319]$), and 30 per-frame bands give -0.672 dB ($[-1.323, 0.086]$). Retaining the true phase at 30 per-frame bands gives only $+0.117$ dB ($[-0.671, 1.084]$). By contrast, keeping all 257 bins but pooling power to 1 s still yields $+0.585$ dB with mixture phase ($[0.285, 1.132]$); the 5-s version yields $+0.249$ dB ($[-0.044, 0.495]$). Thus this interface is especially sensitive to frequency averaging and the grouping rule, while time averaging alone is less destructive. These are sufficiency results for the fixed waveform copy head, not information-theoretic limits or new FiLM results.

C. Main 128-group residual experiment

a) Trained high-resolution Oracle residual separator.: We next test the representation in an actual BS-RoFormer, rather than only a frozen linear copy head. The actual current N_t and mixture x_t are transformed with a 512-point, 128-hop STFT. Mean power in 128 equal-width frequency groups yields time-resolved $\log |B_t|$; the explicit innovation is $R_t = \log |X_t| - \log |B_t|$ on the same grid. These maps are interpolated to the BSR frame grid and projected into its mixture-derived band tokens before axial attention. This zero-initialized residualized $[X_t, B_t, R_t]$ input uses neither FiLM nor a forecaster; its initial waveform output exactly matches ordinary BSR (maximum difference zero).

Both models use the same day-disjoint DCLDE mixtures, transferred backbone initialization, seed, batch order, 6,000 updates, AdamW/cosine schedule, loss, and validation-best checkpoint rule. The existing BSR selected step 5,300; the Oracle model selected step 6,000. On 4,096 matched test examples, BSR scores 9.631 and Oracle-Residual-BSR 16.767 dB target SI-SDRi. The paired improvement is $+7.136$ dB (recording-day 95% interval $[6.970, 7.282]$), with all three held-out days positive. The combined high-resolution prior and explicit innovation interface therefore benefits this controlled separator. Actual N_t is unavailable at deployment; this two-model comparison also does not isolate adapter capacity from the condition information.

b) Causal high-resolution residual conditions.: We then replace the current-background Oracle map with two conditions computed strictly from the preceding 60 s. Each of twelve 5-s history windows has the same 79×128 log-magnitude grid. Annotation-overlapping windows are masked; when one is unavailable, its slot copies the nearest annotation-clean window still inside the past 60 s. This affects 235 of 68,075 eligible current windows. Persistence uses the last clean history map. A lightweight GRU encodes each map into four temporal chunks, processes twelve steps, and decodes a residual correction to persistence. It predicts the entire future 79×128 log-background map, without seeing the current waveform. The predictor is trained for 3,000 updates on training-day clean backgrounds and frozen at its lowest validation log-magnitude MAE. Validation MAE is 0.5456 for the GRU versus 0.5518

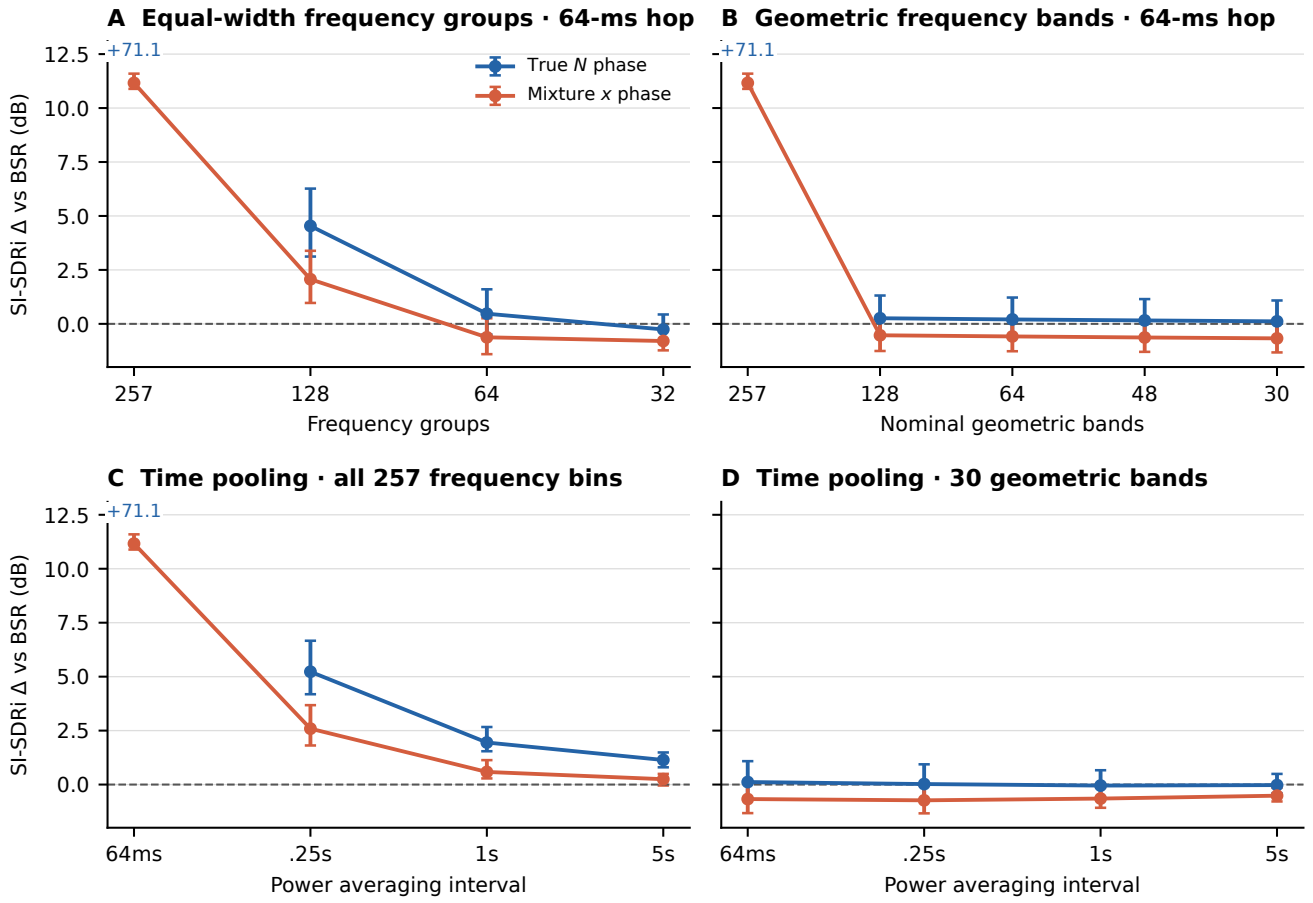


Fig. 2. Current-background Oracle information ladder on identical DCLDE test mixtures. Each point fits the same three-coefficient copy head on training mixtures. Vertical bars are paired 95% day-block intervals for SI-SDRi difference from frozen BSR. Full-resolution true-phase N_t gives +71.1 dB and is above the plotted range; mixture phase is a separate information loss. The threshold depends on frequency grouping and this copy interface.

for persistence (1.12% relative gain). On the held-out test days, the corresponding map MAEs are 0.5308 and 0.5386.

For every condition map B, we form $R = \log|X_t| - B$ and feed $[X_t, B, R]$ through precisely the Oracle residual adapter. The Zero control repeats the fixed training-set mean log background at every frame. The Shuffled control draws an annotation-clean B_j from a *different recording day*: training and testing draw within their respective splits; validation draws from training days because validation has only one day. Persistence takes the last clean history map; Forecast uses the frozen GRU. All four separators start from the same transferred BSR backbone and zero adapter, so their initial waveform output is identical to ordinary BSR. They use identical synthetic events, training-example order, 6,000 update budget, optimizer, loss, validation-best checkpoint rule, and held-out mixtures. Thus the four residual arms differ only in the source of B.

Table III separates interface and background effects. Zero exceeds ordinary BSR by +2.525 dB (recording-day paired 95% CI $[2.262, 2.701]$); Shuffled exceeds BSR by +2.313 dB ($[2.027, 2.905]$). Hence the full BSR-to-causal gain cannot be attributed to correct history alone. The Zero contrast includes both the extra adapter and the current mixture log-spectrum supplied by the innovation path, so it does not separately identify parameter count. With the adapter held fixed, Persistence exceeds Zero by +2.138 dB ($[1.775, 2.377]$) and wrong-day Shuffled by +2.351 dB ($[1.571, 2.661]$), positive on every

TABLE III
MATCHED DCLDE TEST ON 4,096 CLEAN-BACKGROUND MIXTURES. ALL FIVE RESIDUAL ARMS USE THE IDENTICAL 128-GROUP ADAPTER; ZERO USES A FIXED TRAINING MEAN MAP AND SHUFFLED USES ANOTHER RECORDING DAY. ONLY ORACLE RECEIVES THE CURRENT TRUE BACKGROUND. VALUES ARE MEAN TARGET SI-SDRI IN DB.

Condition	SI-SDRi	Gain vs. BSR
BSR (mixture only)	9.631	—
Zero-Residual	12.156	+2.525
Shuffled-Residual	11.944	+2.313
Persistence-Residual	14.295	+4.664
GRU-Forecast-Residual	14.351	+4.720
Oracle-Residual	16.767	+7.136

held-out day. This supports a contribution from temporally relevant historical background beyond architecture and arbitrary background maps. In this original seed, Forecast exceeds Persistence by only $+0.056$ dB ($[0.036, 0.074]$), positive on all three days; the later seed replication in Table IV shows that this small GRU advantage is not stable across training seeds. This seed's Forecast retains 66.1% of the Oracle gain over BSR, with a 2.416 dB Oracle gap ($[2.114, 2.640]$).

D. Three separator seeds on fixed mixtures

We reuse the original run as seed one and independently retrain BSR, Zero, Shuffled, Persistence, and Forecast with two further seeds. All five arms receive the same per-seed transferred backbone initialization, training mixtures, 6,000 updates, 4,096-example validation draw, 100-step validation cadence, and best-validation checkpoint rule. A fixed 4,096-mixture test draw is identical across every arm and seed; sample IDs, SNRs, and mixture SI-SDR are checked before comparison. The background GRU remains the previously selected frozen checkpoint across seeds, so this experiment measures *separator* seed sensitivity, not predictor-training variation.

Correct history exceeds both matched controls for all three seeds and on each of their three held-out recording days. Its advantage over Zero is $2.400 \pm .237$ dB and over wrong-day Shuffled $2.145 \pm .181$ dB. Thus the main historical-information result survives separator initialization variation. Forecast exceeds Persistence in only two of three seeds; its $+.017 \pm .036$ dB average contrast does not support a reliable learned-GRU separation advantage. The original seed's positive day-block interval conditions on that one trained model and should not be interpreted as uncertainty across training seeds.

E. Secondary ablations: time scale and target strength

The time-resolution diagnostic in Table V is an ablation on continuous SanctSound audio, not the main annotation-clean data. Only its 5-s predictor beats persistence on validation (MAE 1.366 versus 1.386 dB); the other scales select persistence. The separate 42-h strength curve also uses unlabeled current background. It shows that forecast conditioning is strength-dependent in that domain and must not be read as a result on annotation-clean current segments.

The harder fixed-5-s experiment on unannotated SanctSound WAV compares Zero-condition FiLM, a current-spectrum oracle, causal forecast, and a time-shuffled condition on 1,152 paired test mixtures. This oracle belongs to the secondary data domain; Table II uses the actual annotation-clean DCLDE N_t. Averaged across training SNRs, the baseline scores 23.470 dB and Forecast-FC-BSR 22.871 dB ($\Delta = -0.599$, CI $[-0.851, -0.346]$); even the oracle is lower. This exposes conditioning/capacity effects that the simpler diagnostic misses. Figure 3 gives the more informative strength sweep. Forecast conditioning helps weak targets by 0.728, 0.643, 0.525, 0.366, and 0.157 dB at $-18, -15, -12, -9, -6$ dB, respectively, then loses from -3 dB onward. The shuffled condition also helps the weakest targets but collapses as SNR rises; therefore weak-target gain alone cannot prove that temporal forecast accuracy, rather than regularization or extra capacity, is responsible.

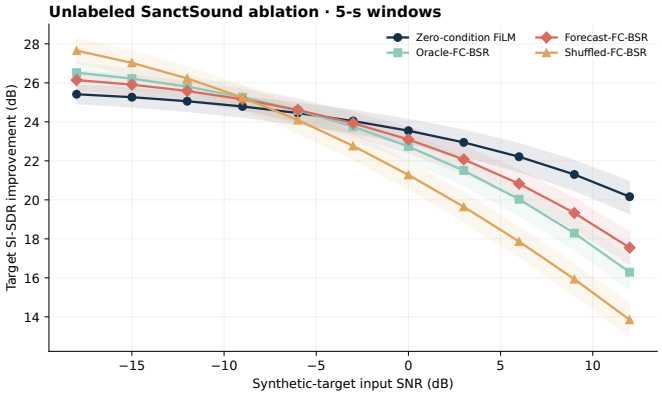


Fig. 3. Ablation on 5-s unlabeled SanctSound background: target-strength sweep. Lines show the four conditions; shaded bands are paired 95% intervals.

IV. Exploratory Real-Event Diagnostics

The following experiments probe external validity and failure modes; they do not replace the exact-stem DCLDE test. Later target and weak-supervision variants reuse the same 103 real calls, so their comparisons are exploratory. The independent new-site check appears in Section V.

A. Frozen-model proxy evaluation

Following learned reference-free metrics [23], [24] and semantic audio embeddings [25], [26], we train, calibrate, and freeze a binary event classifier (test AUROC .9723, AP .9755, F1 .9275). Mixture consistency imposes $\hat{N} = x - \hat{s}$. We restore $\hat{s}$ at gains $\{-60, -30, -24, -18, -12, -6, 0\}$ dB and test whether event probability recovers monotonically. Identity and all-to-event controls reject trivial solutions; changes outside annotated intervals audit preservation.

We evaluate the four frozen 128-group residual models on the exact 103 test calls used by the preliminary FiLM audit. Each call receives twelve strictly earlier annotation-clean maps from the same WAV. Six histories each lack one clean window, filled by the nearest clean window still inside the prior 60 s. Zero uses the training mean, Shuffled a clean background from another test day, Persistence the last historical map, and Forecast the frozen GRU applied to that history. No condition reads the current call. Table VI and Fig. 5 show the results.

Persistence versus Zero lowers $p(\hat{N})$ by .0624 (recording-day paired 95% CI $[-.0682, -.0547]$), raises $p(\hat{s})$ by .0469 ($[.0198, .1006]$), and lowers off-event change by .0255 ($[-.0386, -.0077]$). Forecast shows the same directions versus Zero, with $\Delta p(\hat{N}) = -.0531$ and Δ off-event $= -.0260$. However, wrong-day Shuffled attains lower $p(\hat{N})$ and higher $p(\hat{s})$ than either causal condition: Persistence minus Shuffled is $+.0471$ ($[.0317, .0706]$) for $p(\hat{N})$ and $-.0606$ ($[-.0731, -.0404]$) for $p(\hat{s})$. Persistence protects off-event audio much better ($\Delta = -.0408$, $[-.0519, -.0320]$). Monotonicity is near saturation for all four models and does

TABLE IV

THREE SEPARATOR SEEDS ON THE IDENTICAL 4,096-MIXTURE DCLDE TEST. ENTRIES ARE TARGET SI-SDRI (DB); CONTRAST ROWS ARE PAIRED DIFFERENCES (DB). THE LAST COLUMN IS MEAN ± SAMPLE STANDARD DEVIATION ACROSS SEEDS. ORACLE IS A SINGLE-SEED UPPER BOUND (16.767 DB) AND WAS NOT RETRAINED.

Method or paired contrast	Seed 1	Seed 2	Seed 3	Mean ± SD
BS-RoFormer	9.631	9.662	9.613	$9.635 \pm .025$
Zero-Residual	12.156	10.824	11.553	$11.511 \pm .667$
Shuffled-Residual	11.944	11.355	12.002	$11.767 \pm .358$
Persistence-Residual	14.295	13.425	14.015	$13.912 \pm .444$
GRU-Forecast-Residual	14.351	13.431	14.002	$13.928 \pm .465$
Persistence − Zero	+2.138	+2.601	+2.461	$+2.400 \pm .237$
Persistence − Shuffled	+2.351	+2.070	+2.013	$+2.145 \pm .181$
Forecast − Persistence	+.056	+.005	−.012	$+.017 \pm .036$

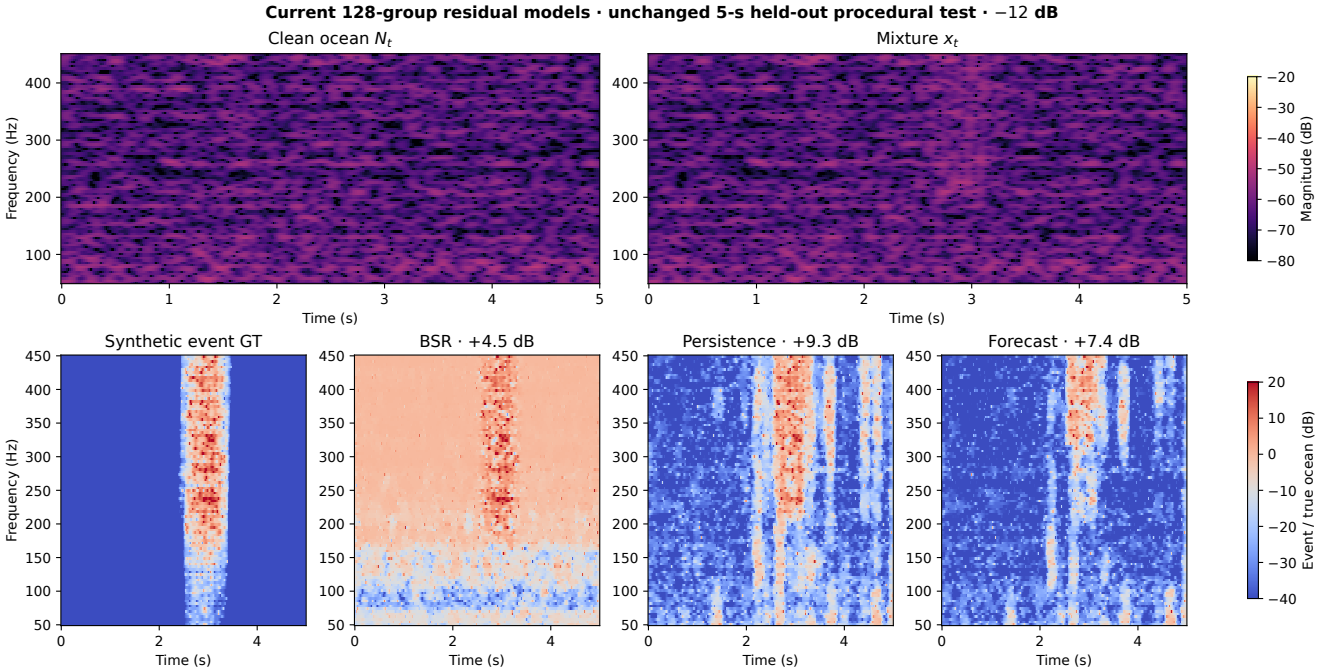


Fig. 4. Current 128-group residual separators on an unchanged held-out −12 dB procedural whale pulse packet, selected by joint median SI-SDRi among 96 fixed examples. The ocean and mixture share an absolute magnitude scale; event stems share a dB scale relative to the true ocean. Compared with BSR, causal Persistence and Forecast suppress much of the off-target ocean, though isolated vertical artifacts remain. Across all 96 examples, the median event/ocean contrast in ground-truth-absent TF bins is −1.37, −31.44, and −36.46 dB for BSR, Persistence, and Forecast. These contrasts measure controlled synthetic leakage, not real-whale stem fidelity.

TABLE V

ABLATION: PREDICTOR-RESOLUTION DIAGNOSTIC ON SANCTSOUND BACKGROUND.

Step	Predictor	Cond.	Base	Gain [95% CI]
5 s	GRU	29.753	28.527	1.226 [1.082,1.381]
30 s	Persist.	29.630	28.534	1.095 [0.935,1.265]
1 min	Persist.	29.661	28.533	1.128 [0.971,1.298]
10 min	Persist.	29.711	28.535	1.176 [1.036,1.326]

Cond./Base/Gain are target SI-SDRi in dB.

not separate them. The classifier probabilities therefore do not reproduce the controlled full-reference ranking; they can reward aggressive extraction, while off-event preservation favors the causal conditions. Isolated real-event stems remain unavailable.

B. Classifier floor and synthetic-target coverage

To audit the high $p(\hat{N})$, we apply the frozen classifier to annotation-clean test audio, truncated to the separator's 9,984 output samples. Its mean probability is .153 on 4,096 day-proportional clean windows and .176 on 103 clean windows from the same WAV, immediately before the corresponding real calls; both medians are .093. The real mixtures average .909. The four separated-background means of .781–.891 are well above the clean floor; paired differences from the same-WAV control range from .606 to .715. This is a classifier-response gap, not a waveform measure of leaked whale energy: artifacts or unmarked events can also elicit responses.

We generate 32 whale-target mixtures per matched clean window (3,296 total), using the five training whale families

TABLE VI

FROZEN 128-GROUP RESIDUAL MODELS ON THE SAME 103 REAL CALLS. ALL MODELS SEE THE SAME MIXTURE; REAL SEPARATED STEMS ARE UNAVAILABLE. OFF-EVENT IS RELATIVE BACKGROUND CHANGE OUTSIDE ANNOTATED CALLS.

Method	$p(x)$	$p(\hat{N})\downarrow$	$p(\hat{s})\uparrow$	Mono.↑	Off-event↓
Zero-Residual	.909	.891	.802	.990	.062
Shuffled-Residual	.909	.781	.910	.989	.078
Persistence-Residual	.909	.829	.849	.992	.037
GRU-Forecast-Residual	.909	.838	.842	.989	.036

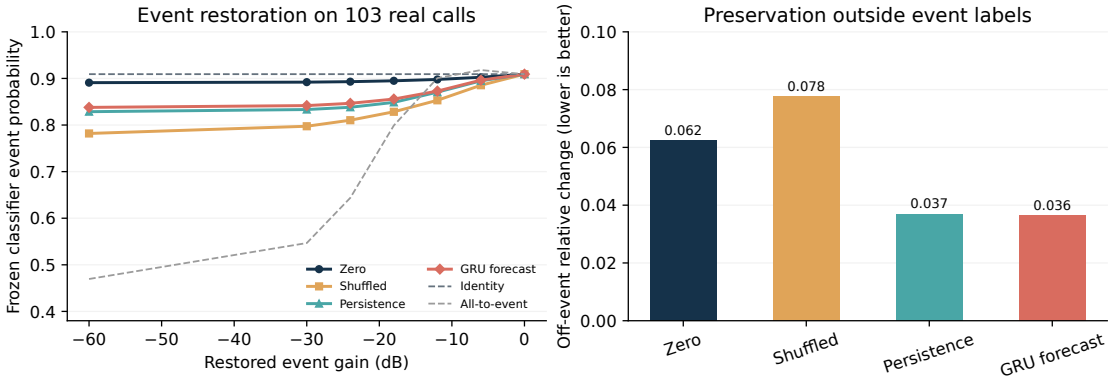


Fig. 5. Frozen-classifier event restoration and off-event preservation for four matched residual models. Identity and all-to-event are degenerate controls. Shuffled scores strongly on event probabilities but changes more audio outside annotated calls than either causal condition.

and $[-18, 12]$ dB sampled strengths. Their mean classifier probability is .407. For a proxy coverage test, each event-bearing clip's frozen 64-D penultimate activation is subtracted from its same-WAV clean activation. After scaling dimensions by synthetic variability, median real-to-real leave-one-out nearest-neighbor distance is .233. Searching all 3,296 synthetic clips gives a real-to-synthetic median of .262, but this reference set is much larger. Using one synthetic clip per real call gives 103 candidates per repeat; across 32 repeats the median distance is .483, or $2.08\times$ the real-to-real distance. A median 51.5% of calls exceed the real-to-real 95th-percentile distance threshold (versus 3.9% with all 3,296 synthetic candidates). Synthetic examples are thus sparse near real events in this *classifier-dependent, equal-size* representation. This motivates target-distribution refinement; neither the embedding nor detector confidence establishes the physical cause of residual event responses without isolated real whale stems. Real background evolution since the preceding clean window also enters this embedding difference.

C. Training-day target calibration and acceptance

We next use only the three training dates to fit target morphology. Of 2,798 annotated call windows, 2,616 have usable same-WAV clean windows before and after the call and sufficient acoustic excess. We interpolate their local background log spectra, take positive mixture-minus-background excess in the annotated time span and 40–950 Hz, apply a smooth mask, and invert with mixture phase to obtain a *pseudo* whale stem. This is an approximate target, not an isolated reference. We estimate call duration, frequency limits and track, sweep slope and curvature, bandwidth, harmonic count and decay, amplitude envelope, jitter, pulse gap, and event count from these training calls. A calibrated procedural route perturbs a sampled whale profile while retaining the original tonal and cavitation vessel generators; the second route randomizes timing and phase of a pseudo-real stem. Each contributes half the new target pool, with vessel draws occupying $4/7$ of the procedural half to preserve the original overall whale/vessel ratio in expectation. The held-out classifier is not used to fit profiles or filter samples.

Under the same 103-call, 32-repeat equal-size coverage protocol, with both new routes represented in every repeat, the median real-to-target nearest distance falls from .483 to .322; the median fraction above the real-to-real 95th-percentile threshold falls from 51.5% to 3.4%. Distance improves in all 32 paired repeats, meeting our pretraining acceptance criterion. The frozen classifier's mean response to new mixtures rises from .407 to .647, but this is an external diagnostic, not an optimization target or proof of stem fidelity.

D. Separation after target-pool revision

We retrain Persistence and Forecast residual separators for 6,000 updates each, with the same transfer initialization and validation selection; the background GRU and event classifier remain frozen. Table VII compares them with their original counterparts. On the same real calls, Persistence lowers $p(\hat{N})$ by .1766 (recording-day paired 95% CI $[-.1920, -.1479]$) and raises $p(\hat{s})$ by .0668 ($[.0414, .1166]$). Forecast changes these probabilities by $-.1129$ and $+.0527$. Both gains co-exist with greater off-event change: $+.0215$ for Persistence ($[.0151, .0324]$) and $+.0068$ for Forecast ($[.0002, .0132]$). New

Persistence extracts more event-like audio than new Forecast, but changes more audio outside the annotation. On the unchanged legacy synthetic test, SI-SDRi falls by 2.100 and 2.620 dB, respectively. The frozen GRU does not improve over persistence after target-pool revision. Retaining vessel draws prevents the severe tonal-vessel regression of an intermediate whale-only pool, yet does not preserve the old whale-proxy scores. Even the new $p(\hat{N})$ values remain far above the matched clean floor of .176. Without isolated real stems, these findings support a target-domain effect on classifier response, not a full-reference real separation gain.

To check optimization separately, we train fixed small sets. With the same loss and backbone, 128 old procedural mixtures rise from 13.85 to 17.20 dB train SI-SDRi in 3,000 updates; 128 empirical mixtures rise from 10.57 to 15.56 dB in 20,000 updates, with little further SI-SDRi gain near the end. Their target-absent event/ocean contrast nevertheless falls from -32.85 to -60.68 dB. This distinguishes remaining target reconstruction error from simple background leakage on these controlled mixtures. We then fine-tune the original strong Persistence checkpoint through 0%, 25%, and 50% pseudo-real stages at one-tenth the original learning rate. Disjoint 80/20 training-day pseudo profiles and fixed old/empirical validation suites select step 6,500 in the 25% stage: validation SI-SDRi changes from $16.26/11.12$ to $16.44/11.84$ dB. Four additional thousand 50%-stage updates further suppress target-absent energy but reduce joint SI-SDRi. The selected checkpoint scores 14.421 dB on the untouched old-synthetic test, yet its real-event $p(\hat{N})/p(\hat{s}) = .819/.823$ fails to retain the $50/50$ model's $.652/.916$. Thus more training helps a fixed set and avoids old-domain forgetting, but alone does not resolve the real-target tradeoff. A matched SI-SDR auxiliary term gains .40 dB on a fixed 128-mixture training probe, yet four thousand full-data 50%-stage updates do not exceed the same dual-validation starting point. We therefore keep the original loss; adding further target-absent suppression is also unsupported by the current controlled leakage audit.

E. Weak supervision from annotated real calls

We next fine-tune the empirical Persistence separator with two train-day synthetic/pseudo-real mixtures and two train-day annotated real calls per update. Only 180 real calls have eligible clean causal histories; 47 validation-day calls select the checkpoint, and the 103 test calls remain untouched until selection. A separate temporal dilated CNN, trained from train-day onset/offset masks, supplies frozen framewise logits C_A; its validation-day frame average precision is .901. It differs in both architecture and training objective from the frozen clip-level test classifier C_B. Neither C_B nor any test-day call enters fine-tuning or checkpoint selection.

For real mixture x, we set $\hat{N} = x - \hat{s}$ exactly and penalize $C_A(\hat{N})$ and $1 - C_A(\hat{s})$ within annotated frames. Outside the annotation we penalize the event-stem RMS relative to the mixture RMS. The supervised half retains the original multiresolution spectral loss. We give the weak term one-tenth weight because the unscaled detector loss rapidly degraded validation SI-SDRi. Selection minimizes validation-day background response subject to event response, off-event alteration, and old-synthetic SI-SDRi constraints fixed before opening the test set. The selected step is 250 of 3,000; later updates fail the SI-SDR constraint.

On the independent C_B and the same 103 calls, $p(\hat{N})$ changes from .652 to .408 (recording-day paired 95% CI for the difference $[-.365, -.093]$) and $p(\hat{s})$ from .916 to .978 ($[.044, .072]$). Off-event relative change is .058 versus .061; its paired CI $[-.004, .008]$ includes zero. The mean matched-clean probability is .176. The per-call clean-normalized logit residue $\operatorname{logit} p(\hat{N}) - \operatorname{logit} p(N_{\text{clean,matched}})$ falls from 3.606 to 1.389. On the unchanged 4,096-mixture synthetic test, however, SI-SDRi falls from 12.195 to 10.323 dB. These independent-detector results support partial transfer from real event annotations, but do not reach the clean-background response or establish waveform-level recovery of an unobserved real stem.

To limit forgetting, we freeze all band encoding and attention layers and train only the residual projection and mask output head. The original 50/50 Persistence model is a frozen teacher. The new objective retains supervised synthetic/pseudo-real loss, adds two-stem L_1 teacher consistency, and matches $C_A(\hat{N})$ to C_A on the immediately preceding annotation-clean 5-s window from the same WAV. Event retention and off-event preservation remain in the real loss. Specifically, $L = L_{\text{GT}} + .01L_{\text{real}} + .1L_{\text{teacher}}$, where $L_{\text{real}} = .25L_{\text{match}} + .1L_{\text{event}} + 2L_{\text{off}}$; L_{match} is a frame-masked smooth-L_1 loss on training-detector logits. Of 414,658 parameters, 99,774 representation parameters stay frozen; all 241 frozen state tensors are bit-identical to the teacher in the selected checkpoint. Validation-day constraints select step 900. A stronger teacher-weight trial does not materially improve the validation clean-response gap, so it is not promoted to test.

On the reused 103-call test cohort, independent C_B gives $p(\hat{N})/p(\hat{s}) = .461/.960$, versus $.652/.916$ before weak tuning and $.408/.978$ after full-network weak tuning. Off-event change is .064, versus .058 before tuning; its paired increase is positive across the three dates. Crucially, unchanged old-synthetic SI-SDRi is 12.156 dB, only .038 dB below the 50/50 teacher and 1.833 dB above full-network weak tuning. The clean-normalized logit residue is 1.917 (teacher: 3.606; full weak: 1.389). This is a more conservative Pareto point, but it does not reach .25–.30 background event probability or the matched-clean response of .176. Because this cohort was already reported for the preceding model, this follow-up comparison is exploratory rather than a fresh blind test.

V. Independent New-Site Evaluation

A. One-shot real-event test at a new site

We froze the original-seed BSR, four original residual arms, both 50/50-target residual arms, the adapter-only weak checkpoint, the background GRU, and the evaluation detector before opening a geographically distinct test set. The Gulf of St. Lawrence (GSL) continuous recordings and call-center annotations were collected in 2015–2017; none entered DCLDE

TABLE VII
SAME FROZEN REAL-CALL EVALUATION AFTER TARGET-POOL REVISION, AND SI-SDRI ON THE UNCHANGED OLD SYNTHETIC TEST. CURRICULUM SELECTS ON TWO SEPARATE VALIDATION SUITES; THE 128-GROUP INTERFACE AND PREDICTOR ARE FIXED.

Condition	Training targets	Old-synth SI-SDRi (dB)	Real $p(\hat{N})\downarrow$	Real $p(\hat{s})\uparrow$	Off-event$\downarrow$
Persistence	Original procedural	14.295	.829	.849	.037
Persistence	50/50 empirical	12.195	.652	.916	.058
Persistence	50/50 + real weak	10.323	.408	.978	.061
Persistence	Frozen trunk + clean-match teacher	12.156	.461	.960	.064
Persistence	Curriculum (25% pseudo)	14.421	.819	.823	.037
GRU Forecast	Original procedural	14.351	.838	.842	.036
GRU Forecast	50/50 empirical	11.731	.725	.895	.043

target-profile calibration, weak training, or model selection [27], [28]. To bound storage and rule out outcome-based selection, we fixed SHA-256 filename ranks 13–28 among 50 continuous files. The first 12 ranks were excluded from confirmatory analysis after a preprocessing audit found that their initial evaluation had omitted the per-clip mean subtraction used in all DCLDE training. No model weight, hyperparameter, or detector was changed. The untouched final eight hours contain 386 annotated upcalls; 373 permit a full strictly past 60-s history. Each source WAV is resampled from 32,768 to 2,000 Hz and each 5-s clip is demeaned as in DCLDE. History windows with call centers within a 2-s guard are replaced by the nearest annotation-negative window in that past minute. Shuffled uses a different selected file because per-file recording dates are not available. GSL marks call centers, not onsets and offsets, so the off-event region uses a declared ± 1.5-s center proxy. It is less precise than the original DCLDE interval mask, and the matched annotation-negative background is not an isolated clean stem. Detector probabilities may be miscalibrated after this site change, so we compare stems within the same files and report the local negative-window floor. The center-proxy off-event measure is not numerically interchangeable with the DCLDE onset/offset measure.

Table VIII shows a consistent but qualified new-site effect. Persistence lowers background event response relative to Shuffled by $.064$ with either event or file weighting; the file-block paired 95% interval is $[-.124, -.013]$. At the same time, file-equal event-stem response is .009 lower ($[-.013, -.004]$), and off-event relative alteration $.016$ higher ($[.007, .027]$). Forecast is effectively tied with Persistence here. Both causal arms suppress background detector response far more than Zero, but alter the off-center mixture by about $.54$ relative RMS, and $p(\hat{N}) = .361$ remains above the local annotation-negative floor of .193. The 50/50 and weakly adapted models do not give a clear Pareto improvement. Thus correct history transfers to the new site as a background-response advantage over wrong-file conditioning, while event retention and background preservation worsen. We do not interpret lower $p(\hat{N})$ alone as robust real-stem separation. This external test uses one frozen separator seed; the three-seed analysis above addresses only controlled DCLDE mixtures.

VI. DISCUSSION AND CONCLUSION

The preliminary 30-band FiLM model does not gain even from the true current 30-band background. A successful full-waveform copy check and the fixed-head information ladder show why this negative result cannot rule out useful background information: compression and injection remain confounded. With a 128-group residual interface, Oracle gains 7.136 dB over BSR; past-60-s Persistence and Forecast retain 4.664 and 4.720 dB. Zero and wrong-day Shuffled also gain 2.525 and 2.313 dB, so the expanded interface contributes. Within that interface, correct history adds 2.138 dB over Zero and 2.351 dB over Shuffled in the original seed. Three-seed replication preserves these contrasts at $2.400 \pm .237$ and $2.145 \pm .181$ dB, positive for all nine seed-day combinations. The original $.056$-dB GRU gain shrinks to $+.017 \pm .036$ dB across seeds and reverses sign in one seed. The learned forecast has no established separation advantage over short-term persistence; the original seed remains 2.416 dB below Oracle.

On real calls, the original four-condition ranking differs between frozen classifier probabilities and off-event preservation. The clean-background floor does not explain the high $p(\hat{N})$, and an equal-size embedding audit finds sparse synthetic coverage near real calls. Training-day empirical targets close much of that measured gap and improve both real-call event probabilities, while increasing off-event changes and lowering old-domain SI-SDRi. The frozen GRU does not beat persistence after this target revision. Independent-detector weak fine-tuning lowers the background event proxy further while preserving the event-stem proxy and approximately maintaining off-event audio, but costs another 1.872 dB on the unchanged synthetic test. Freezing the separator representation and adding teacher consistency nearly eliminates that controlled-domain loss, at the expense of weaker real event suppression and a small off-event increase. Thus proxy improvement does not establish a uniform real separation gain. Reuse of three test days, uncertain event labels, and unavailable isolated stems further limit inference.

The one-shot GSL cohort tests a new recording site without tuning on its calls. Its lower background-stem detector response with correct history is encouraging, but the simultaneous loss in event-stem response and off-event preservation rules out a broad real-domain victory. These proxies depend on an imperfect cross-site detector and center-only annotations. The existing 30-band FiLM frontend permits inspection and playback, but

TABLE VIII

ONE-SHOT GSL NEW-SITE EVALUATION ON 373 ANNOTATED UPCALLS FROM 13 EVENT-BEARING FILES. THE FROZEN DCLDE-TRAINED DETECTOR GIVES $p(x) = .907$ AND $p(N_{\text{ann.}-\text{neg.}}) = .193$ ON A PRECEDING ANNOTATION-NEGATIVE WINDOW. LOWER $p(\hat{N})$ AND OFF-EVENT ALTERATION ARE DESIRABLE; HIGHER $p(\hat{s})$ AND RESTORATION MONOTONICITY ARE DESIRABLE. CENTER-BASED MASKS AND NO ISOLATED REAL STEMS MAKE THESE SEMANTIC/PRESERVATION PROXIES.

Frozen separator	$p(\hat{N}) \downarrow$	$p(\hat{s}) \uparrow$	Monotonic ↑	Off-event ↓
BS-RoFormer	.688	.931	.970	.489
Zero-Residual	.894	.631	.993	.126
Shuffled-Residual	.425	.936	.964	.528
Persistence-Residual	.361	.927	.955	.543
GRU-Forecast-Residual	.361	.927	.954	.543
50/50 Persistence	.439	.915	.954	.557
50/50 GRU Forecast	.447	.912	.952	.569
Frozen-trunk weak Persistence	.439	.915	.955	.557

the 128-group model has not been deployed there. Future work requires stronger isolated real-event references and better background preservation during target adaptation. Classifier restoration does not measure waveform fidelity.

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